



SDGs Governance: Silo Implementation by Government Agencies in Tackling Sustainable Development Goal 2 (Zero Hunger) in Kenya

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Abstract

The pursuit of Sustainable Development Goal 2 (Zero Hunger) in Kenya has been constrained by siloed governance, where ministries, agencies, and counties implement food security programs in isolation. This study examines the structural and institutional factors sustaining these silos and their implications for food and nutrition security. Drawing on policy reviews, key informant interviews across 23 purposively selected counties, and county-level case studies, the research finds that fragmented coordination leads to duplication of programs, inefficient resource use, weak accountability, and limited progress in addressing hunger and malnutrition, particularly in arid and semi-arid lands. Using the Institutional Analysis and Development (IAD) framework, the study shows how bureaucratic traditions, resource competition, donor-driven vertical projects, and devolution-related gaps reinforce siloed implementation. The paper proposes a three-pronged integrated governance framework focusing on empowered coordination structures, aligned financing and incentives, and interoperable data systems.

Acknowledgement

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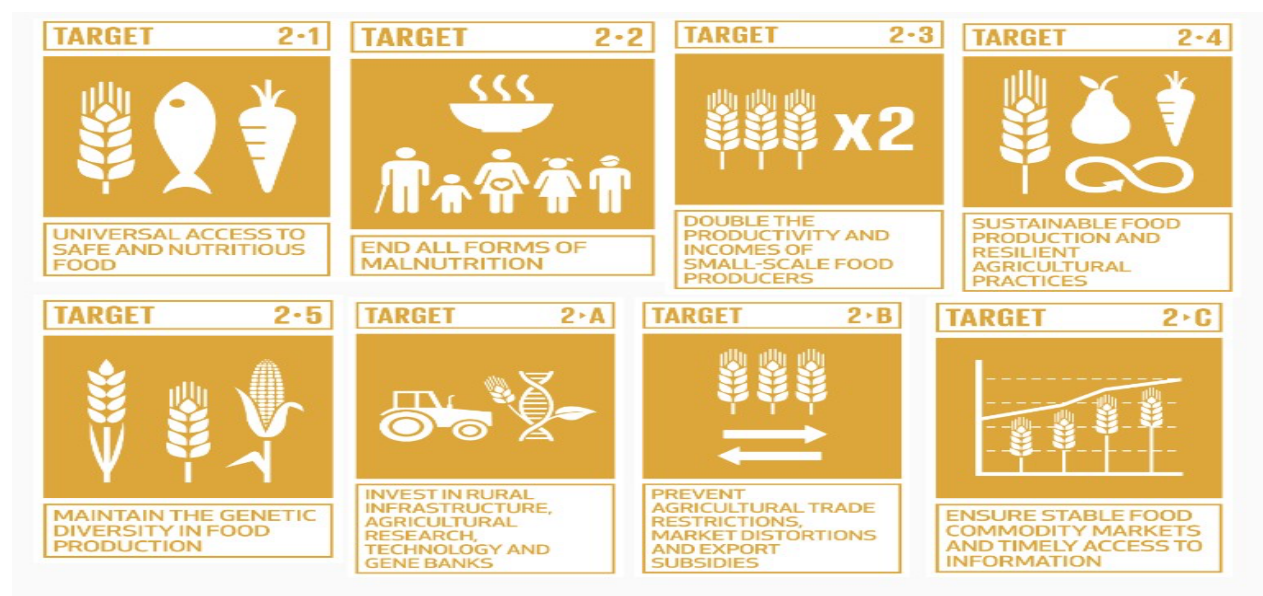
List of Acronyms

Acronym	Full Form
ASALs	Arid and Semi-Arid Lands
CASSCOM	County Agriculture Sector Coordinating Committee
CBOs	Community-Based Organizations
FCDC	Frontier Counties Development Council
FNS	Food and Nutrition Security
IAD	Institutional Analysis and Development
KIIs	Key Informant Interviews
MDAs	Ministries, Departments, and Agencies
MoALD	Ministry of Agriculture and Livestock Development
MoH	Ministry of Health
CASSCOM	County Agriculture Sector Steering Committee
JASSCOM	National Joint Agricultural Sector Steering Committee
NDMA	National Drought Management Authority
SDGs	Sustainable Development Goals
VNR	Voluntary National Review
WoG	Whole-of-Government

Introduction

1.1 Background

The Sustainable Development Goals (SDGs) represent a global framework aimed at ending poverty, protecting the planet, and ensuring prosperity for all. SDG 2 specifically focuses on achieving Zero Hunger by 2030 through sustainable agricultural practices, improved nutrition, and resilient food systems. In Kenya, however, implementation has been persistently fragmented: government ministries, NGOs, and private sector actors address specific aspects of food security without comprehensive coordination. This lack of integration undermines the broader objectives of SDG 2.



SDG 2 Targets and Indicators

Pic Courtesy: Thomas Okanagan Tourism Association

Kenya's food security situation is deteriorating. The 2024 Voluntary National Review (2024 VNR) reports that the prevalence of moderate or severe food insecurity rose from 60.3 per cent in 2020 to 62.4 per cent in 2022, while severe food insecurity nearly doubled from 17.0 per cent to 30.5 per cent over the same period. Rural populations bear the heaviest burden at 63.4 per cent, compared to 52.2 per cent in urban areas. Climate change compounds these pressures: Kenya's agricultural sector, which contributes approximately 22.5 per cent of GDP (2025 Economic Survey) is increasingly threatened by rising temperatures, erratic rainfall, prolonged droughts, and flooding, all of which directly undermine progress on SDG 2.

SDG 2 is one of the most interconnected goals, linking directly to SDG 1 (No Poverty), SDG 3 (Good Health), SDG 5 (Gender Equality), SDG 13 (Climate Action), SDG 14 (Life Below Water), SDG 15 (Life on Land), SDG 16 (Strong Institutions), and SDG 17 (Partnerships). Addressing

hunger and malnutrition therefore demands an integrated, multi-sectoral governance approach precisely what siloed implementation prevents.

To effectively achieve SDG 2 therefore requires a multi-sectoral approach. According to (FAO et al., 2023), governments including in Kenya play a central role in strengthening food systems through policy formulation, investment in rural infrastructure, regulation, and the promotion of equitable access to agricultural services, food security programmes, and climate adaptation initiatives

The private sector complements the government effort through mobilising investment, driving innovation, and delivering market-based solutions across agricultural value chains (World Economic Forum, 2022). Agribusinesses, financial institutions, technology firms, and food processors support productivity, value addition, access to finance and markets, and the adoption of climate-smart and digital agricultural solutions, particularly for smallholder farmers (CGIAR, 2021; GSMA, 2023). However, fragmented and siloed implementation in the sector continue to limit coordination, create policy inconsistencies, weaken value chain integration, and discourage long-term private sector investment (African Development Bank, 2022; AGRA, 2023).

Small-scale food producers remain central to Kenya's food security and agrifood systems; however, they continue to face structural constraints including low productivity, limited access to markets, finance, and technology, and persistent income disparities, with female headed farmers disproportionately affected (HLPE, 2013; IFAD, 2023; FAO, 2022). These constraints reduce their capacity to contribute effectively to sustainable food systems and national food security. Addressing these disparities requires coordinated and inclusive policies that improve access to resources, enhance productivity, strengthen resilience, and promote gender equality within agrifood systems (World Bank, 2023).

1.2 Problem Statement

Despite Kenya's efforts to achieve SDG 2, persistent challenges related to hunger, malnutrition, and food insecurity remain widespread. The key issue is the siloed governance structures in the implementation of food security programs, which result in fragmented efforts, inefficiencies, and duplication of resources. The absence of coordination between agriculture, health, social protection, and environmental sectors hinders sustainable outcomes, particularly for vulnerable populations in arid and semi-arid regions of the country. Despite interventions, silo approach persists.

1.3 Objectives

- Assess the current siloed governance structure in SDG 2 implementation in Kenya and its impacts on food security outcomes.
- Map key stakeholders involved in food security governance at national and county levels.
- Identify challenges and opportunities for multi-sectoral collaboration.
- Develop an integrated governance framework to enhance cross-sectoral collaboration, policy coherence, and efficient resource use.

1.4 Research Questions

- How do governance silos manifest within the institutional architecture responsible for SDG 2 implementation?
- What are the specific negative impacts of these silos on food security and nutrition outcomes?
- What existing mechanisms exist for integration, and what explains their success or failure?
- What strategic interventions can create a more effective governance model for achieving SDG 2?

1.5 Scope and Limitations

The study covers national and county government institutions, with deep-dive examples from contrasting counties, and sampled private sector and civil society stakeholders. The analysis covers 2015 (SDG adoption) to 2024, with reference to earlier policies where relevant. The study is primarily qualitative, and findings from 23 selected counties, while purposively representative, are not statistically generalizable to all 47 counties.

Literature Review

2.1 Theoretical Framework: Understanding Silo Governance

In Kenya's Food and Nutrition Security (FNS) context, silo implementation refers to the disconnected, parallel, and uncoordinated execution of programs by different government ministries, departments, agencies, and counties, each focusing narrowly on its sectoral mandate without synergistic alignment. As noted in the National Information Platform for Food Security and Nutrition, 'breaking down the silos that separate nutrition sector players is a key challenge in overcoming inconsistencies and promoting cross-sectoral synergies' (GoK/NIPFN, 2021, p. 23).

The theoretical roots of silo formation trace to Max Weber's ideal-type bureaucracy, which emphasizes hierarchy, specialization, and formal rules. While designed for efficiency and neutrality, this model creates vertical structures that inhibit horizontal information flows. Three theoretical perspectives are particularly relevant to the Kenyan context. New Public Management (NPM) reforms of the 1980s and 1990s introduced agency autonomy and performance contracts that incentivized inward-looking departmental behavior at the expense of cross-agency collaboration (Christensen and Lagreid, 2007). Resource Dependency Theory holds that organizations hoard information, budgets, and expertise to maximize autonomy, creating barriers to inter-agency cooperation (Pfeffer and Salancik, 1978). Institutional Theory explains how siloed practices become deeply entrenched 'rules of the game,' with path dependency making integrative reforms politically and institutionally difficult (North, 1990).

In the Kenyan and African context, these generic challenges are compounded by the colonial inheritance of administrative structures designed for control rather than integrated service delivery (Mamdani, 1996), by donor-driven vertical projects that create parallel management structures accountable to funders rather than national systems (OECD, 2011), and by the 2010 constitutional devolution that created new vertical silos between national and county governments alongside existing horizontal ones (Cheeseman, Lynch, and Willis, 2016).

Devolution (the 2010 Constitution) mandated county Governments to undertake several functions including the Agriculture and Livestock, Health, water and Sanitation, Early childhood education and Land management system. In implementing the SDG 2 on food nutrition security, they are the front runners.

2.2 Empirical Evidence on Silo Governance in Kenya and Africa

The empirical literature on silo implementation in Kenya's food and nutrition security sector is consistent and compelling. Mwangi, Bhardwaj, and Karanja (2021) provide the most rigorous institutional analysis, finding that despite the overarching Food and Nutrition Security Policy, implementation is crippled by weak inter-ministerial coordination, conflicting mandates, and resource competition. Their case studies show that agricultural extension services may successfully increase maize yields without contributing to reductions in micronutrient deficiencies a direct consequence of the Ministry of Agriculture and Ministry of Health operating in isolation. Gillespie, van den Bold, and Hodge (2019) corroborate these findings from a broader Sub-Saharan African perspective, concluding that siloed sectoral policies are a fundamental barrier to nutrition improvement.

On vertical silos introduced by devolution, Chege, Kimiywe, and Ndungu (2015) conducted an early empirical case study in Machakos County, documenting significant coordination gaps between national and county entities: weak technical capacity at county level, unclear role delineation, and inadequate information-sharing mechanisms. Owuor and Kioli (2017) extend this analysis, showing county agricultural programs misaligned with national food reserve management, producing situations where local surpluses co-existed with deficits in neighboring regions. Mose and Ochieng (2021) specifically examine devolution and SDG 2, emphasizing how county-level silo challenges directly undermine Zero Hunger targets.

In nutrition programming, Nguyen et al. (2017) find that nutrition-specific interventions such as micronutrient supplementation led by health agencies are routinely delivered entirely separately from nutrition-sensitive programs like agricultural diversification, leading to sub-optimal outcomes despite significant investment. Pomeroy-Stevens, D'Agostino, and Adero (2016) use network analysis to empirically map these silos in Kenya's nutrition sector, confirming that cross-sectoral links between health, agriculture, and water actors are structurally weak, not merely perceptual. Korir et al. found that improvements in nutrition governance between 2012 and 2023 were driven by stronger political commitment and policy frameworks, although significant challenges persisted in financing, accountability, and intersectoral coordination (Korir et al., 2025). Another study further revealed that ministries and agencies often implemented nutrition-related programmes independently, with fragmented monitoring systems and weak joint accountability mechanisms limiting effective collaboration (Korir et al., 2025). Similarly, research on Kenya's nutrition governance landscape established that sectors such as health, agriculture, education, and water continued to plan and budget separately, leading to duplication of interventions and weak institutional integration (Public Health Nutrition, 2024).

For smallholder farmers, the consequences are concrete. A Tegemeo Institute (2020) case study demonstrates that high-yield seed promotion by the Ministry of Agriculture failed in one region because a complementary irrigation project from the Water Ministry was delayed by separate budgeting cycles. Research on extension services in East Africa finds that farmers receiving coordinated support on seeds, fertilizer, and credit were 40 per cent more likely to fully adopt improved soil fertility management practices (Journal of Agricultural Education and Extension, 2020). The FAO (2021) similarly documents that distributing drought-tolerant seeds without concurrent water harvesting initiatives yields minimal results during droughts, a pattern repeatedly observed in Kenya's ASAL counties.

Globally, the theoretical responses to siloed governance, the Whole-of-Government (WoG) approach, Joined-Up Government (JUG), and digital integration platforms are well-articulated (Christensen and Lagreid, 2007; Ling, 2002; Janowski, 2015), but their implementation in devolved, resource-constrained African governance systems remains a significant challenge that this study examines empirically. One finding on this review is that there is significant research gap in understanding the political economy of silo-busting in contexts like in Kenya: Who wins and who loses from integration? How are powerful vested interests that benefit from silos overcome?

2.3 Kenya's Institutional Landscape for SDG 2

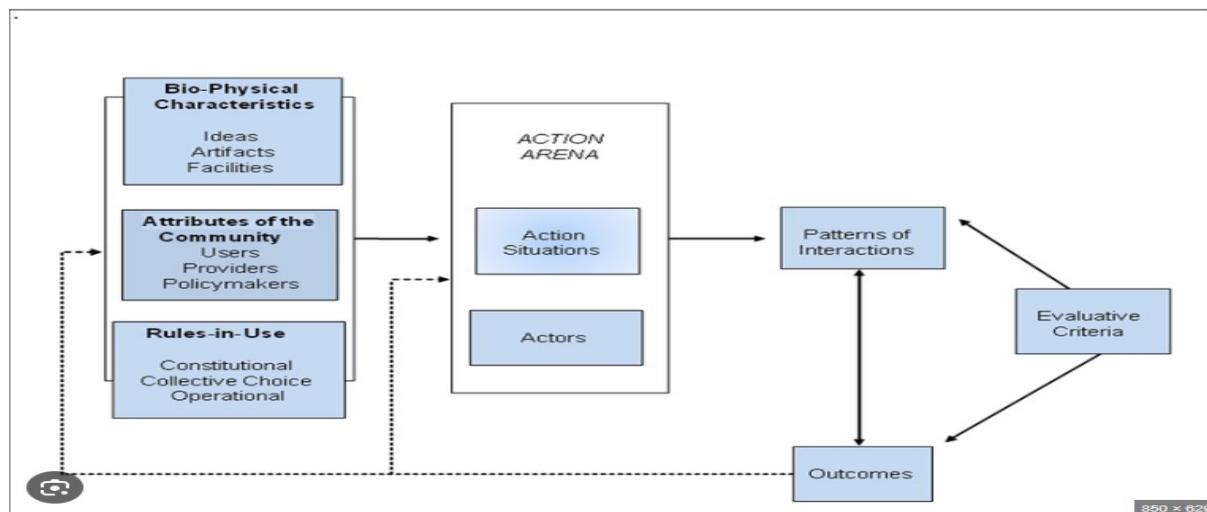
The following table maps the principal institutions responsible for SDG 2 implementation in Kenya, illustrating the multi-actor landscape within which silo governance operates.

Level	Institution / Actor	Key SDG 2 Role
National	MoALD (lead ministry)	Agricultural policy, food security, input regulation
National	Ministry of Health	Nutrition, maternal and child health, malnutrition treatment
National	NDMA	Drought risk management, ASAL early warning systems
National	National Treasury	Budget allocation, SDG integration, VNR reporting
National	SAGAs (KALRO, NIA, AFC, NCPB)	Research, irrigation, credit, grain reserves
County (47)	County Agriculture & Health Departments	Devolved implementation, extension, community services
Parliament	National Assembly, Senate	Legislation, budget approval
Development Partners	FAO, WFP, USAID, EU, AfDB, GIZ	Finance, technical support, program implementation
Private Sector	Agri-processors, input firms, markets	Production, processing, distribution, value chains
Research Institutions	KALRO, ILRI, Universities, ARIN	Evidence generation, policy recommendations
Civil Society / CBOs	SDG Kenya Forum, cooperatives, farmer groups	Community advocacy, local delivery

2.4 The IAD Framework as Analytical Lens

This study applies Elinor Ostrom's Institutional Analysis and Development (IAD) framework to diagnose Kenya's SDG 2 governance challenges. The IAD framework centres on the action arena comprising actors and action situations shaped by exogenous variables including rules-in-use, biophysical conditions, and community attributes (Ostrom, 2005). It enables systematic diagnosis of institutional performance without prescribing one-size-fits-all solutions (Ostrom, 2011), and has been applied to agricultural governance (McGinnis and Ostrom, 2014), natural resource management (Clement, 2010), and policy implementation failures (Garcia-Reyes, 2016) in contexts directly analogous to Kenya's.

Below is the imagery of the IAD Framework



(Ostrom, 2005, p. 15)

Research Methodology

3.1 Research Design

This research adopted a qualitative, exploratory multi-case study design, selected for its suitability to address how and why questions about complex institutional phenomena. The study proceeded in two phases: a comprehensive document review covering national food security strategies, budget statements, SDG Voluntary National Reviews, and reports from agencies including FAO, WFP, and IFAD; and an empirical field study involving key informant interviews (KIs), structured questionnaires, and focus group discussions.

3.2 Sampling Strategy and County Selection

Purposive sampling guided selection of both counties and individual participants which is the standard approach in qualitative research for identifying information-rich sources relevant to the research questions. The 23 counties were selected based on three explicit criteria: (1) geographic diversity across Kenya's distinct agro-ecological zones, ensuring representation of high-potential agricultural counties (e.g., Nakuru, Nandi, Bungoma), pastoralist ASAL counties (e.g., Marsabit, Mandera, Turkana, Tana River, Samburu, West Pokot), and mixed livelihood counties (e.g., Kitui, Makueni, Baringo);

(2) variation in SDG 2 indicators, including food insecurity rates, stunting prevalence, and agricultural productivity, to capture contexts ranging from food-secure to chronically food-insecure; and (3) evidence of differing approaches to cross-sectoral coordination, enabling comparative analysis of what works and what does not. The ASAL-heavy FCD counties (Marsabit, Mandera, Samburu, Turkana, Tana River) were deliberately included given their concentration of the most food-insecure populations.

The full sample comprised 18 national government officials drawn from the ministries and SAGAs of National Treasury, Health, Agriculture, Lands, Water and Irrigation, Gender, Environment, Cooperatives, and Interior; 23 county governments; 6 staff from international agencies including UN bodies (UNDP, FAO, UNEP), USAID, EU, Danish and Finnish Embassies, the African Union, AfDB, and GIZ; 8 research institutions (KALRO, University of Nairobi, JOOUST, Kenya Marine and Fisheries Institute, Tea Research Institute, ILRI, and AERC); and 9 representatives from civil society, including the SDG Kenya Forum, youth groups in sustainable aquaculture, and community groups in selected counties. Recruitment continued until theoretical saturation was reached.

3.3 Data Collection

Semi-structured KIs used an open-ended guide that ensured consistency while allowing flexibility to probe emerging themes. Structured questionnaires focused on meeting frequency, coordination platform effectiveness, silo impacts, and stakeholder interactions. Focus group discussions were convened with homogenous groups of 2 to 4 participants (mid-level officials from different ministries; local NGO representatives) to surface shared experiences and collective norms, particularly within the private sector and CBOs.

3.4 Data Analysis and Coding Process

All KI transcripts and questionnaire responses were entered into a structured Excel workbook for systematic management. Thematic analysis was conducted in two stages. In the first stage, two coders (the primary researcher and a research assistant) independently read all transcripts and applied initial open codes capturing governance patterns, coordination mechanisms, silo manifestations, and stakeholder interactions. Codes were derived inductively from the data and deductively from the IAD framework's key constructs (action situations, actors, rules-in-use, patterns of interaction, and outcomes). In the second stage, the two coders compared and reconciled their coding, resolving disagreements through discussion. Themes were then identified by grouping related codes into higher-order categories. Intercoder reliability was informally assessed through discussion until consensus was reached on all major themes. The resulting themes were interpreted against the IAD framework and existing literature, with the narrative findings supported by direct participant quotations. Quantitative questionnaire data was visualised in charts (Figures 4.1- 4.9) to triangulate and illustrate qualitative findings.

3.5 Ethical Considerations

The research adhered to high ethical standards throughout. All participants provided informed consent, were assured of confidentiality, and participation was entirely voluntary. No identifying information about individual respondents is disclosed in this report.

Findings and Discussion

4.1 National Government

The National Government of Kenya bears the primary mandate for SDG 2 policy direction, with several ministries including Health, Education, Agriculture, National Treasury, Social Protection, Lands, and Environment playing critical implementation roles. Despite this multi-agency architecture, findings from interviews across these ministries reveal a governance landscape characterised by fragmented coordination, limited joint resourcing, and institutional silos.

Inter-ministerial engagement does occur, with meetings convening on a largely quarterly basis and bringing together the key ministries responsible for SDG 2-related mandates. These forums extend beyond government, incorporating external stakeholders such as development partners, Parliament, farmers' cooperatives, the private sector, and academia. While this breadth of participation reflects an acknowledgement of the multi-sectoral nature of food and nutrition security, the periodicity and informality of these engagements suggest they function more as information-sharing forums than as strategic coordination mechanisms capable of driving joint programming or resolving inter-agency tensions in implementation.

The most significant structural gap is the near-absence of formalised, sector-wide engagement platforms for FNS coordination. The National Joint Agricultural Sector Steering Committee (JASSCOM) serves a narrow agricultural mandate, leaving ministries in adjacent domains health, social protection, education without an equivalent home for multi-sectoral dialogue. Budgetary fragmentation mirrors this structural gap: programme budgets remain specific to individual state departments, with no provision for joint budgeting across ministries. This fiscal ring-fencing means that even where policy intent acknowledges the interconnectedness of food security and nutrition, the financial architecture does not enable collaborative resource deployment.

Despite these systemic constraints, respondents identified instances of successful coordination. The Border Management Protocols and the Multi-Annual National Control Plan (MANCP) in fisheries regulation were cited as models of structured inter-agency collaboration that have demonstrated genuine integration in practice.

Vignette 1: Duplication of Seed Distribution Programs

One of the clearest illustrations of duplication emerged from ASALs county Turkana, where both the Ministry of Agriculture and a donor-funded nutrition program operating through the Ministry of Health distributed seed packages to the same households in the same planting season. The agriculture program distributed drought-tolerant maize seeds; the nutrition program distributed diversified seed packages including legumes and sweet potato vines. Neither program was aware of the other's activities, no combined distribution list existed, and the seed types were not complementary since the maize distributed competed for the same small plots as the nutrition program crops. One county agriculture officer described the situation:

"We only found out when farmers told us they had received two sets of seeds, Maize and sweet potatoes, and didn't know what to plant when. There was no registry, no coordination meeting, each program had its own targets and reporting lines. We were both serving the same household, spending double the resources, and the farmer was confused." County Agriculture Officer (Anonymized)

This was repeatedly observed in programmes including National Fertilizer Subsidy Programme, National accelerated Agricultural Input Access programme (NAAIAP), National Value Chain support programme and County fertilizer subsidy programmes which targeted similar households.

This illustrates how the absence of shared beneficiary registries and joint planning, both structural consequences of siloed budgeting and mandates, directly reduces efficiency and community trust.

Vignette 2: Successful Coordination Fisheries Regulation

The Border Management Protocols and the Border Control and Operations Coordination Committee (BCOCC) for fisheries regulation provides a contrasting example of successful multi-agency coordination. Developed jointly by the Kenya Coast Guard Service (KCGS), State Department for Fisheries, Kenya Fisheries Service, KRA, Kenya Bureau of Standards, and county governments, the plan established shared inspection protocols, harmonized data collection, and joint enforcement operations covering Lake Victoria and coastal fisheries.

A key informant from the Kenya Fisheries Service described the impact:

"Before the BCOCC, every agency had its own inspection checklist and reporting format. Fishers were being inspected multiple times by different officers with different standards. Now we go together, one form, shared data. Compliance has improved and the countries actually trust the national process because they helped design it." Senior Officer, Kenya Fisheries Service

Some other cited successful coordination in the sector included National Agricultural Value Chain Development Project implemented by Government with support from World Bank.

This demonstrates that integrated coordination is institutionally feasible in the Kenyan governance context. The enabling factors shared problem recognition, joint design, harmonized data systems, and embedded county participation provide a replicable template for other sectors.

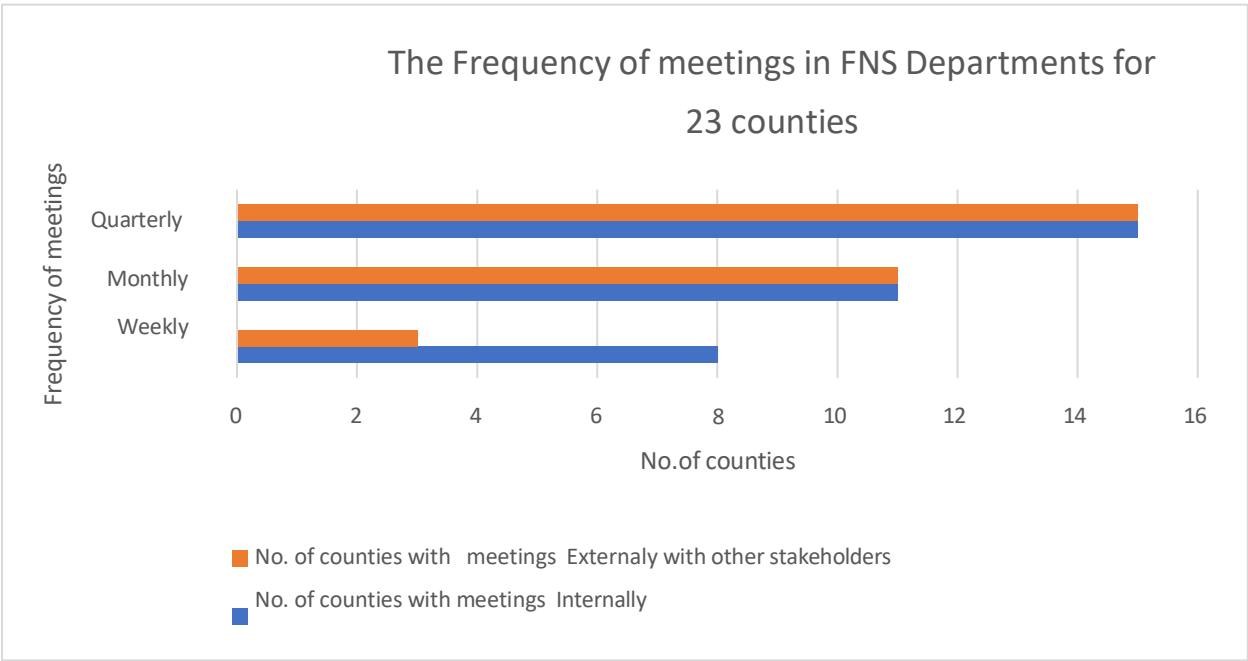
4.2 County Governments

The 23 counties surveyed were selected to represent Kenya's geographic, agro-ecological, and governance diversity. All surveyed counties have Agriculture, Livestock, Food Security, and Blue Economy under one executive department headed by a County Executive Committee Member. However, coordination within and across county departments remains fragmented.

More than 65 per cent of surveyed counties hold internal meetings quarterly, while 50 per cent additionally hold monthly meetings and few hold weekly meetings. External meetings with stakeholders occur quarterly in over 60 per cent of counties with ASAL counties in the FCDC

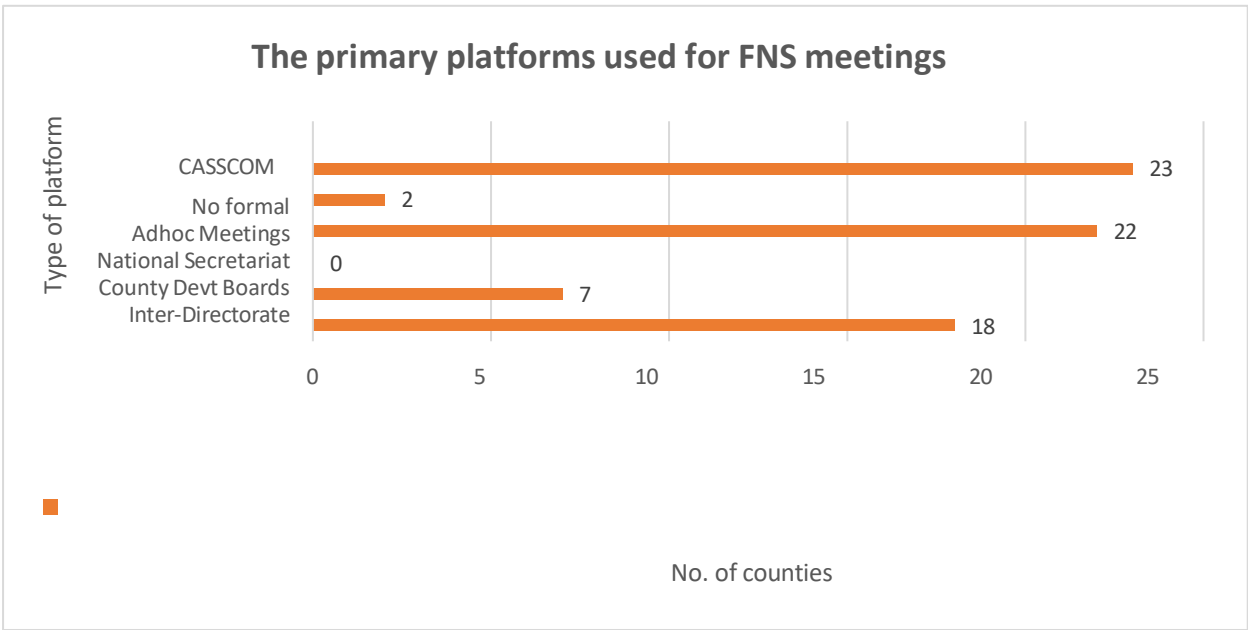
cluster (Marsabit, Mandera, Samburu, Turkana, Tana River) reporting more frequent weekly engagements, likely driven by the urgency of food security challenges in those areas.

Figure 4.1: Chart of the frequency of meetings



All counties had established CASSCOM platforms, though most had enabling bills still awaiting formalization in county assemblies. Over 90 per cent reported holding ad hoc meetings, and more than 78 per cent held inter-directorate meetings reflecting a pattern of reactive, unstructured coordination rather than systematic joint planning.

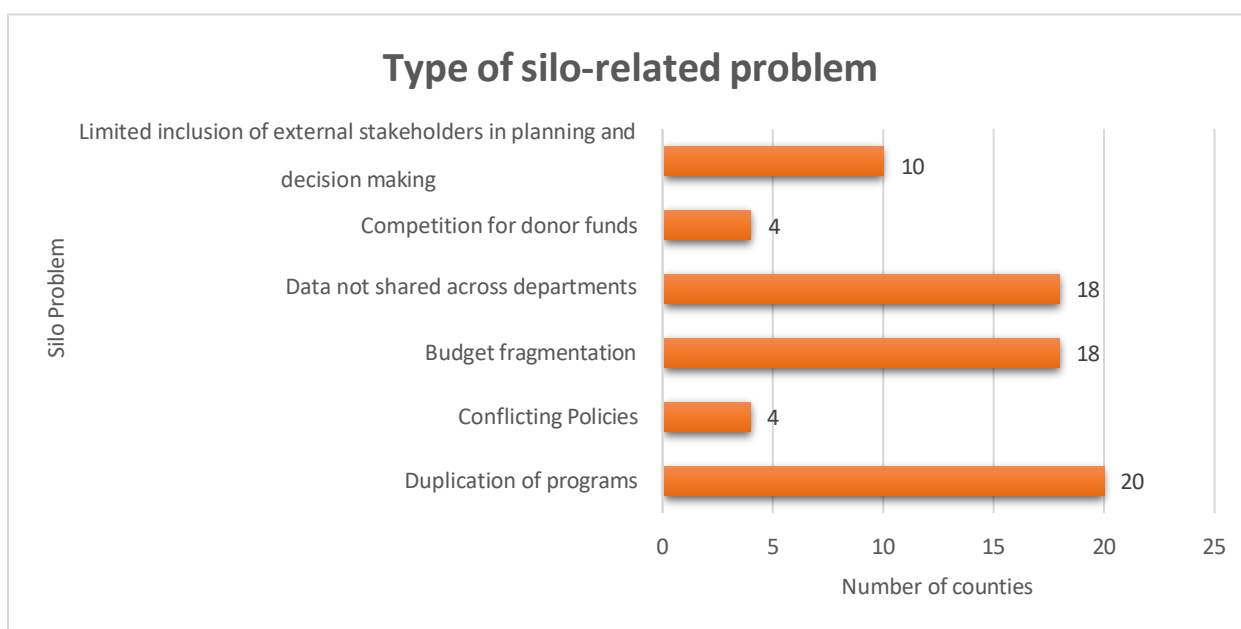
Figure 4.2: Platforms used in Meetings



	Inter-Directorate	County Devt Boards	National Secretariat	Adhoc Meetings	No formal	CASSCOM	
No. of counties	18	7	0	22	2	23	

The quantitative findings are clear on silo impacts: 87 per cent of county respondents reported that silo governance leads to duplication of programs the highest-cited effect. A further 78 per cent cited data hoarding or non-sharing, and budget fragmentation equally. Delayed services and uneven distribution of resources were each cited by 34 per cent as the primary impacts on vulnerable groups.

Figure 4.3: Impacts of silo approaches as observed



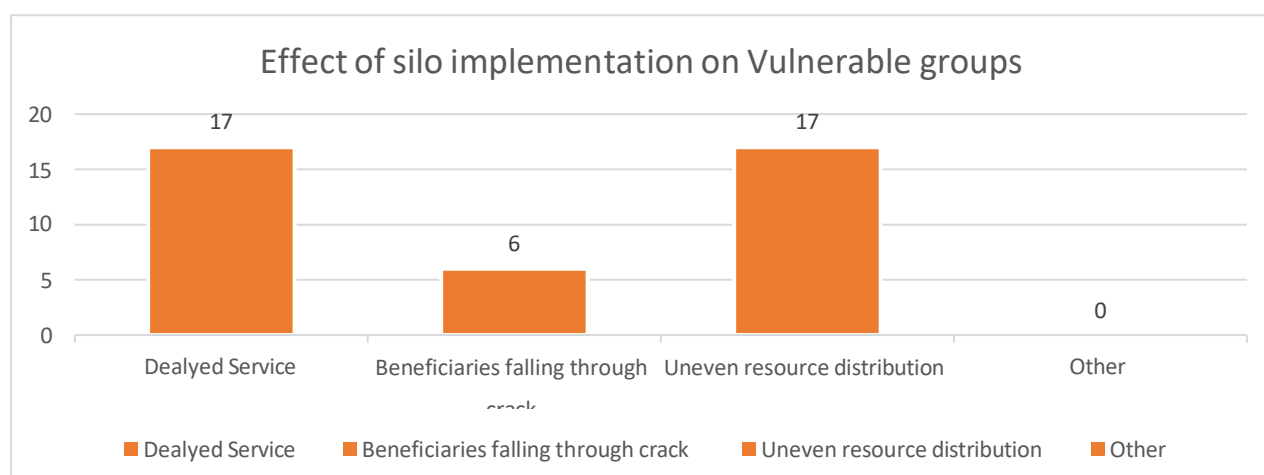
Context 3: Fragmentation Leaving Vulnerable Households Behind

In a focus group discussion with mid-level county officials in a high-potential agricultural county revealed the human cost of budget fragmentation:

"We had a program for fertilizer subsidy and another for climate change support running at the same time in the same ward. The farmer had to visit three different offices agriculture, Environment, and the NGAO's office to get each component. One female small-scale farmer (anonymized) missed the climate support entirely because she couldn't attend the meeting at the Environment office on the same day as the agriculture training. These programs should have been one program, one window." Mid-level County Official, Focus Group Discussion.

Counties like Muranga were running the modest Digital farmer registry with linked programmes like Inua Mkulima. This ran parallel to some national government programmes with similar objectives. The replication was observed across counties including Kakamega, Kwale, Migori and ASAL counties. This illustrates how administrative silo structures translate directly into exclusion of vulnerable households particularly women with mobility or time constraints from the full benefit of programs ostensibly designed to reach them.

Figure 4.4 Impact of silo approach on the vulnerable groups

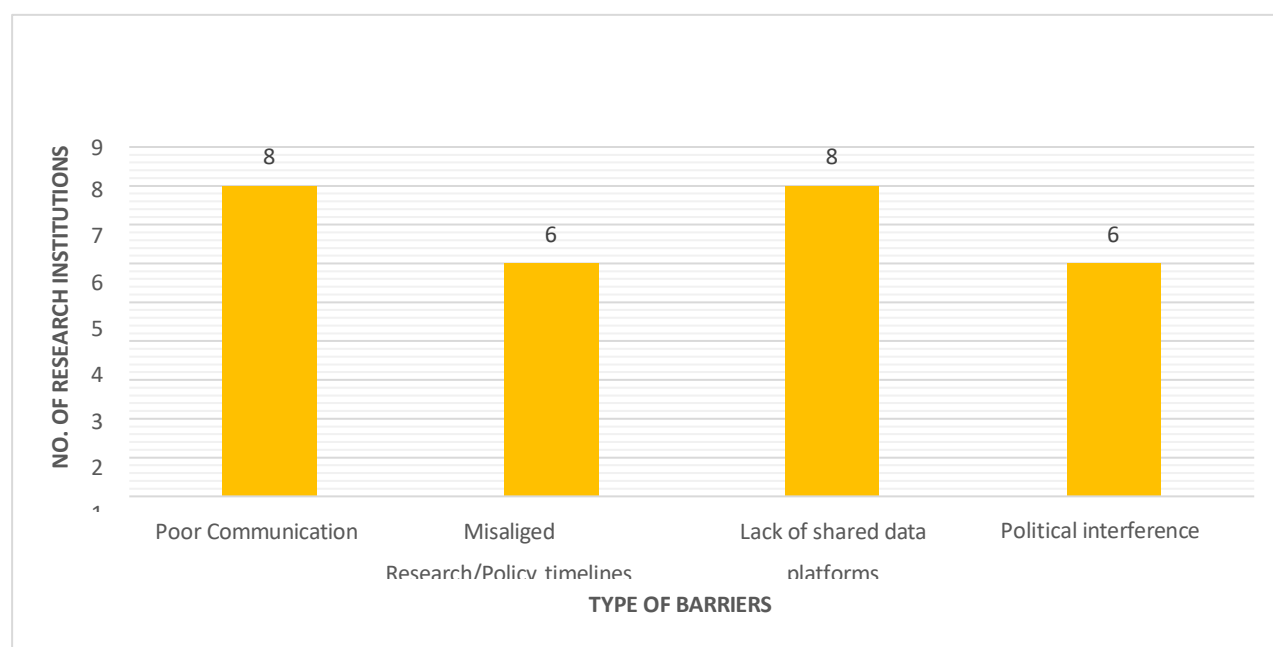


4.3 Research Institutions

Research institutions are central to overcoming silo approaches in SDG 2 implementation. The study engaged 8 research institutions 5 government-affiliated (KALRO, University of Nairobi, JOOUST, Kenya Marine and Fisheries Institute, Tea Research Institute) and 3 non-governmental (ILRI, AERC, ARIN).

Poor communication and lack of shared data platforms were cited as the biggest barriers to linking research to policy, followed closely by misaligned research priorities and political interference.

Figure 4.5: Types of barriers to linking research findings to policy



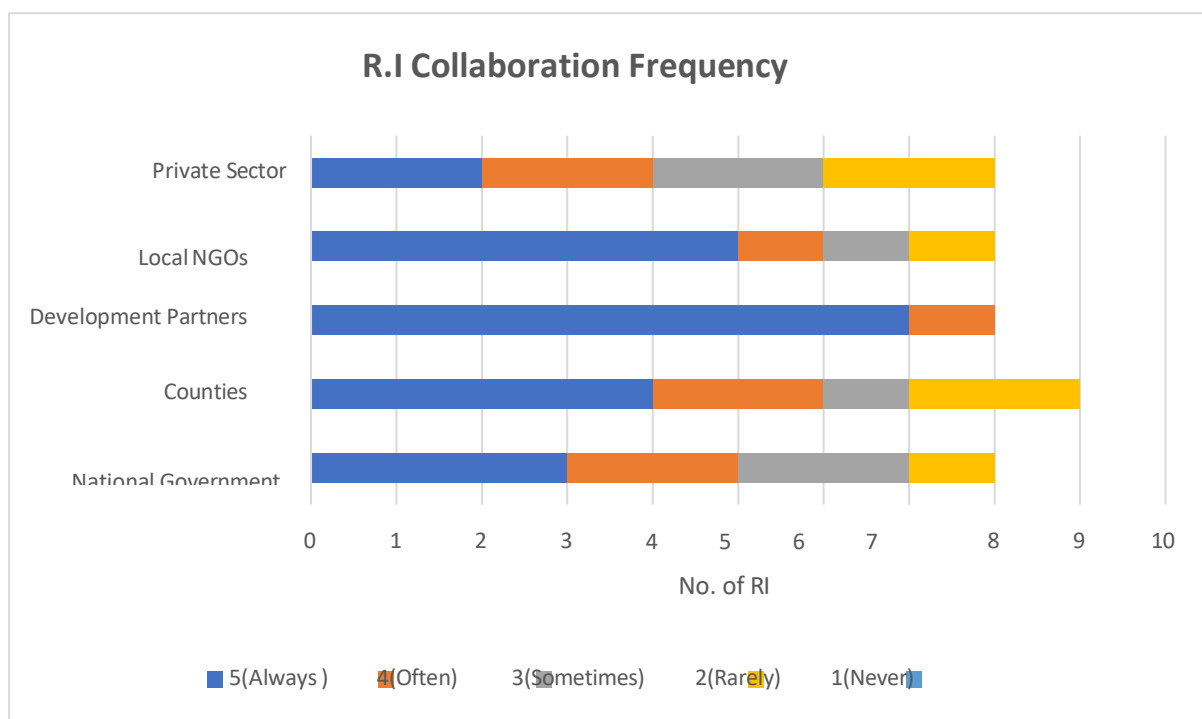
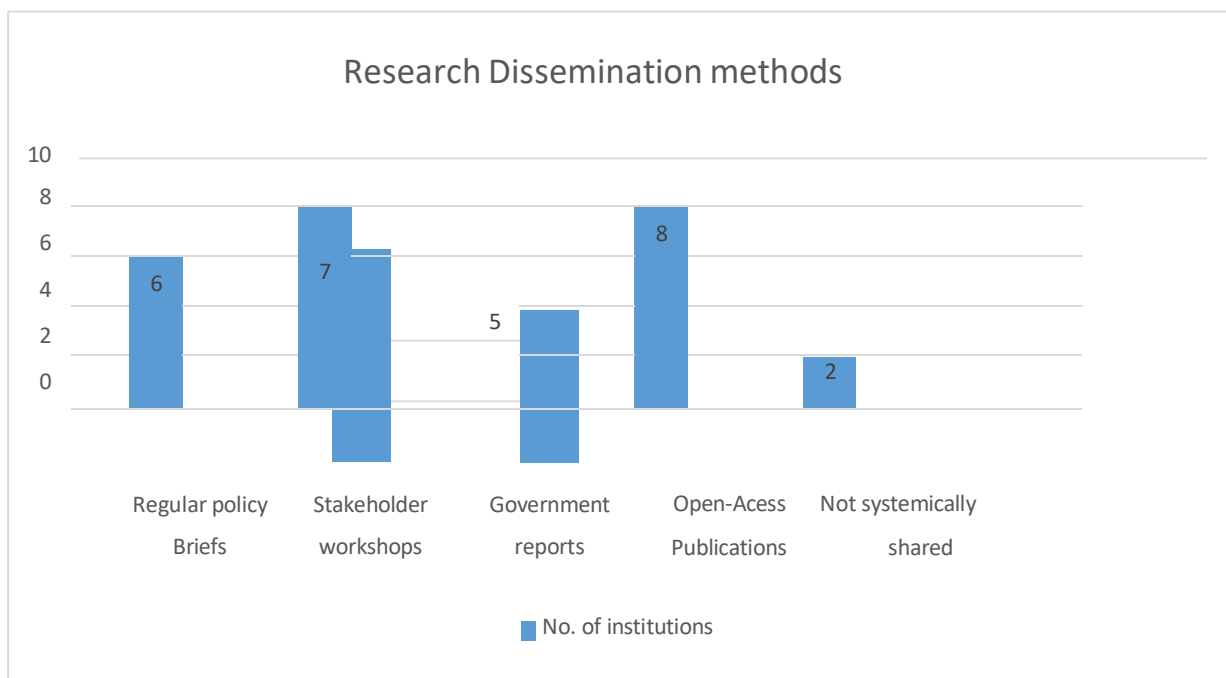


Figure 4.6: The level of frequency in collaborating by Research institutions

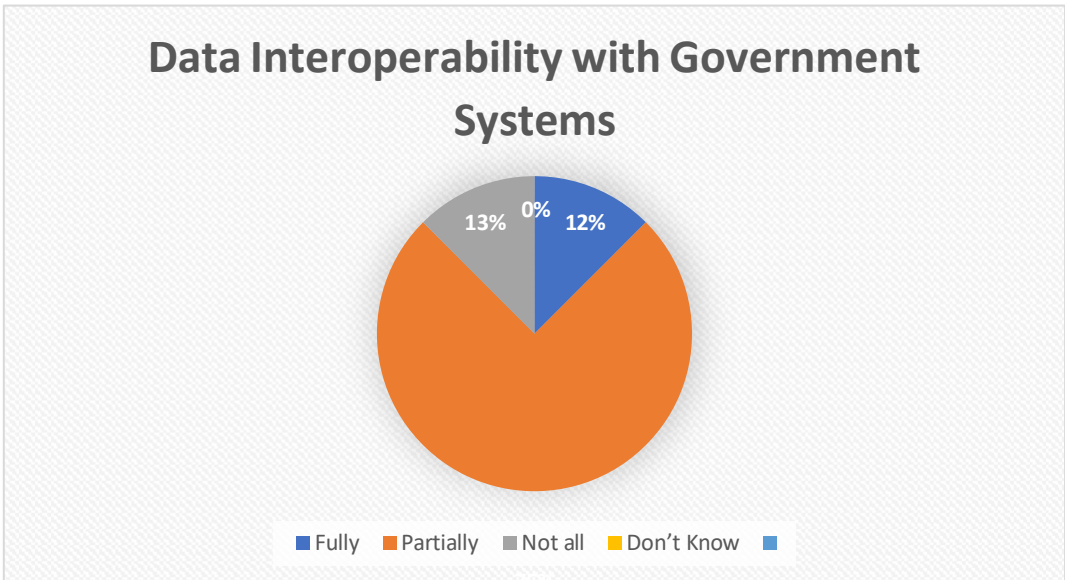
Collaboration was found to be substantially more frequent with development partners and local NGOs than with government and the private sector a troubling finding given that government is the primary target for policy influence.

Figure4.7: Ways in which research findings are disseminated



On dissemination, open-access platforms ranked highest (over 29 per cent of responses), followed by stakeholder workshops and government reports; more than 7 per cent reported that findings were not systematically shared at all.

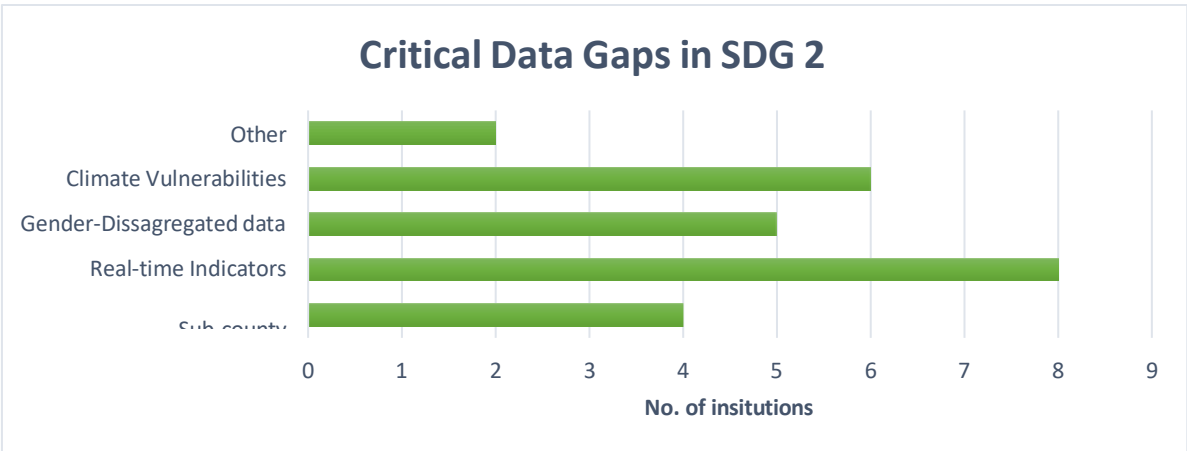
Figure 4.8: The status of data interoperability with government systems



More than 75 per cent reported that data interoperability with government systems was partial, with only 12 per cent achieving full interoperability.

The most critical data gaps cited were: lack of real-time food security indicators, limited tracking of climate vulnerabilities, and insufficient gender-disaggregated data for comprehensive SDG 2 reporting.

Figure 4.9: The critical data gap areas on SDG 2



4.4 Development Partners

Development partners were unequivocal in their support for a multi-sectoral approach to SDG 2 implementation, consistently affirming that the breadth and complexity of food and nutrition security demands coordinated, cross-sectoral interventions rather than siloed programming. Among the agencies consulted, FAO and WFP placed particular emphasis on the importance of strengthening national food systems, framing this work within the broader one-health approach, which recognises the interdependence of human, animal, and environmental health in achieving sustainable food security outcomes.

Despite this shared commitment to multi-sectoralism, development partners identified several systemic barriers that undermine joint implementation in practice. Chief among these were perforated and insufficiently accountable government systems, which were cited as a major hindrance to the coherent delivery of programmes across agencies. Closely related to this was the issue of disjointed funding mechanisms within government budgeting processes, where the absence of harmonised financial planning prevents resources from being aligned to shared programmatic objectives.

A further challenge flagged by development partners was the insufficiency of data and the lack of data interoperability across agencies. This gap is particularly acute in the Frontier Counties Development Council (FCDC) counties, where the number of people facing hunger and climate-related disaster cannot be reliably ascertained. The dependence on Kenya National Bureau of Statistics (KNBS) surveys to generate such data introduces significant delays in the identification of need and the deployment of appropriate responses, with direct implications for the timeliness and effectiveness of humanitarian and development interventions.

Development partners also highlighted the limited interaction between the private sector and government as a structural weakness in the food systems landscape. Areas including market access, produce quality, agricultural diversification, and mechanisation were identified as domains where collaboration between government and the private sector remains underdeveloped, constraining the potential for market-driven solutions to food security challenges.

A particularly urgent concern raised by development partners especially AfDB was the shrinking and increasingly volatile global donor funding landscape. The sector has historically depended heavily on external donor support for life-saving interventions, and the contraction of this support poses significant risks to programme continuity. The closure of USAID programming was cited as a concrete illustration of this vulnerability, having left several ongoing interventions without a clear funding pathway. Partners cautioned that without a robust multi-sectoral coordination framework and stronger domestic financing, programmes risk collapse in the absence of sustained donor commitment, underscoring the need to build systems that are resilient to fluctuations in external support.

4.5 Civil Society, CBOs, and Smallholder Farmers

Engagement with civil society organisations, community-based organisations, and smallholder farmers revealed a starkly different coordination experience from that observed at the national government level. At the community level, interaction is frequent and organic, with many CBOs reporting daily engagement among their members. Many of these organisations operate across

multiple counties, particularly in areas with shared agro-ecological or socio-economic characteristics, reflecting an informal but functional approach to geographic coordination.

A notable structural gap emerged with respect to farmers' cooperatives, which were found to be well-established and relatively effective in the cash crop sub-sector, yet almost entirely absent in the food crops sub-sector. This disparity has significant implications for policy influence and representation, as smallholder food crop farmers lack the organised voice that cooperatives would otherwise provide in government and policy forums. As a result, the concerns and priorities of this constituency remain underrepresented in the planning processes that most directly affect their livelihoods.

Interviewees from this group consistently observed that both national and county governments have tended to focus narrowly on the production dimension of agriculture, with insufficient attention paid to the broader value chain. Post-harvest handling, processing, marketing, and distribution were identified as areas where government engagement is limited, leaving farmers to navigate complex market dynamics with little institutional support beyond the farm gate.

In terms of inclusivity, the findings paint a more encouraging picture at the community level. Meetings convened by CBOs and smallholder farmer groups were reported to be broadly inclusive, with deliberate efforts made to incorporate youth, persons with disabilities, and women in decision-making processes. This stands in contrast to the more hierarchical and sometimes exclusionary dynamics observed in formal government coordination structures, and suggests that community-level platforms offer a model of participatory governance that could inform more inclusive national coordination approaches.

In the FDC counties, a distinctive pattern emerged whereby CBOs and smallholder farmers reported substantially more interaction with development partners than with either national or county government agencies. This dynamic points to a significant gap in government outreach and service delivery in these marginalised regions, and suggests that development partners have, by default, assumed a coordination and support role that would more appropriately be led by government institutions.

Finally, community-based actors identified several key barriers to realising the full potential of multi-sectoral collaboration at the local level. These included limited access to partners who could support agricultural mechanisation and the adoption of new technologies, as well as the constraining effect of outdated land policies that restrict investment, tenure security, and the productive use of land. Addressing these barriers will require sustained engagement not only with agricultural ministries but also with the Ministries of Lands and Environment, reinforcing the case for the integrated, multi-sectoral governance frameworks that this study set out to examine.

4.6 SDG Interlinkages in Practice: Trade-offs and Synergies

A recurring finding across stakeholder groups is that SDG 2 implementation does not occur in isolation from other goals, yet the governance system treats it as if it does. Trade-offs identified included in areas of Budget control, artificial boundaries, data hoarding and Accountability. However, existing synergies were observed in areas of JASSCOM/CASSCOM, regional economic blocs, some donor programme stakeholder engagement and emerging data systems.

The following examples from the data illustrate how trade-offs and synergies manifest in relevant SDGs:

Trade-offs: SDG 2 versus SDG 6, 13, 14 and 15 (Life on Land): Several county respondents and development partners noted that fertilizer subsidy programs promoted by the Ministry of Agriculture (aligned with SDG 2 production targets) conflict directly with soil and biodiversity conservation objectives (SDG 6, 13, 14 and 15). In Uasin Gishu (Rift Valley County) and Bungoma County (Western) for example, the intensive promotion of mono-cropped maize with subsidized chemical fertilizer (DAP and CAN) has contributed to documented soil acidification over five years, a trade-off that was never surfaced in any joint planning forum between the agriculture and environment departments. The absence of a shared performance framework means there is no mechanism to identify, flag, or resolve such contradictions.

Synergies: SDG 2 and SDG 1 and 3 (Good Health and Well-being): The most frequently cited example of successful synergy was Kenya's school meals program, in which the Ministry of Education, MoH, and MoALD have progressively integrated activities. In counties where the program has achieved strongest coherence, linking local sourcing from smallholder farmers to health-screened nutritional content to teacher training, outcomes improved across social protection, SDG 2 (child nutrition) and SDG 3 and 4 (health and school) simultaneously. Respondents attributed this to a joint technical working group with representation from all three ministries, CBOs and county governments that meets frequently and has a shared monitoring framework. This model, however, remains the exception rather than the rule across Kenya's food security architecture.

Proposed Framework for Integrated Governance

Drawing from the findings of fragmented, silo-based SDG 2 implementation, this study recommends a three-pillar integrated governance framework targeting the core dysfunction areas of coordination, finance, and information.

Recommendation 1: Reinvent Governance and Coordination Structures

The findings revealed weak, advisory-only inter-ministerial arrangements with no binding mandate to enforce collaboration. It is recommended that an empowered National Food Systems body be established, chaired at Cabinet level and comprising Cabinet Secretaries from Agriculture, Health, Finance, and Environment with real decision-making authority. Coordination hubs should be replicated at sub-national level, creating platforms for local government officers, NGOs, and community representatives to jointly plan and monitor progress. Existing regional blocs (FCDC, LREB, SEKEB) should be formally integrated into this architecture, given their demonstrated capacity for inter-county coordination in ASAL contexts.

Recommendation 2: Align Financing and Incentives

Financial flows currently reward sectoral competition rather than collaboration. It is recommended that SDG 2 budget tagging be introduced across ministries and that Integrated Financing Frameworks pool resources for multi-year, cross-sectoral programmes. Joint funding windows should require inter-ministerial proposals as a condition of access.

Recommendation 3: Integrate Data and Knowledge Systems

The proliferation of sector-specific, non-interoperable data systems undermines evidence-based decision-making at every level. It is recommended that a National Food Security Data Hub be established as a centralised, open-access platform integrating agricultural, health, climate, and market data using common standards.

These three recommendations are interdependent and should be implemented concurrently. Stronger coordination structures drive better financing alignment, and is guided by integrated data, creating a self-reinforcing cycle toward coherent SDG 2 governance.

Conclusion

Kenya's progress on SDG 2 is significantly impeded by siloed governance at both national and county levels. Ministries, departments, agencies, and counties operate with limited coordination, producing duplication of programs, fragmented budgets, and inefficiency problems consistently confirmed across all stakeholder groups in this study. Counties, though central to food and nutrition security delivery, rely on irregular and ad hoc coordination mechanisms that weaken inclusion of vulnerable groups. Research institutions face barriers of poor data interoperability and weak policy linkages. The private sector and smallholder farmers remain constrained by inconsistent policies and fragmented value chains. Coordination forums such as CASSCOM and JASSCOM provide dialogue platforms but lack the authority and resources to dismantle entrenched silos.

These challenges are not insurmountable. The fisheries BCOCC example demonstrates that integrated inter-agency coordination is institutionally feasible in the Kenyan context. The school meals program shows that SDG 2 and SDG 3 synergies can be captured when a joint technical working group has a shared mandate and monitoring framework. What distinguishes these successes from the prevailing silo pattern is not capacity but institutional design binding authority, shared budgets, and common data systems.

The three-pillar integrated governance framework proposed in Chapter Five offers a path from fragmented to coherent SDG 2 implementation. Priority actions include: empowering NASSCOM/CASSCOM with fiscal authority linked to the budget process; establishing a permanent technical secretariat for CASSCOM with a joint monitoring mandate; creating joint national-county implementation units for specific SDG 2 projects with pooled funding; aligning CIDPs to include a single costed, multi-sectoral food and nutrition security chapter; developing a National Food Security Data Hub; and reforming civil service performance contracts to include cross-sectoral collaboration targets.

The fundamental insight of this study, consistent with the IAD framework diagnosis, is that Kenya government has correctly identified the silo problem and established formal coordination mechanisms but those mechanisms function as dialogue forums rather than decision-making bodies because the core incentives of budgets and political accountability remain locked within siloed structures. Dismantling silos therefore requires changing not just organizational charts, but the fiscal and accountability architecture within which those organizations operate.

References

- Chege, P. M., Kimiywe, J. O., and Ndungu, Z. W. (2015). Impact of devolution on nutrition governance in Kenya: A case study of Machakos County. *Journal of Agriculture, Food Systems, and Community Development*, 5(4), 77–89.
- Cheeseman, N., Lynch, G., and Willis, J. (2016). Decentralisation in Kenya: The governance of governors. *Journal of Modern African Studies*, 54(1), 1–35.
- Christensen, T., and Lagreid, P. (2007). The whole-of-government approach to public sector reform. *Public Administration Review*, 67(6), 1059–1066.
- Clement, F. (2010). Analysing decentralised natural resource governance: Proposition for a politicised institutional analysis and development framework. *Policy Sciences*, 43(2), 129–156.
- FAO. (2013). *FAO strategy for partnerships with the private sector*. FAO.
- FAO. (2021). *Integrated climate risk management for sustainable agriculture in the Horn of Africa: Case study report*. FAO.
- Garcia-Reyes, M. (2016). Institutional analysis of agricultural subsidies: The case of Agro Ingreso Seguro in Colombia. *Estudios Politicos*, 49, 45–68.
- Gillespie, S., van den Bold, M., and Hodge, J. (2019). Nutrition and the governance of agri-food systems in South Asia and sub-Saharan Africa: A systematic review. *Food Security*, 11(2), 287–303.
- GoK/NIPFN. (2021). *Review of policies on food security and nutrition and the use of evidence in improving human nutrition in Kenya*.
- Government of Kenya. (2018). *Evaluation report of the One Million Kitchen Gardens Initiative*. Ministry of Agriculture.
- Journal of Agricultural Education and Extension. (2020). Barriers to adoption of integrated soil fertility management in East Africa, 26(2), 153–170.
- Mamdani, M. (1996). *Citizen and subject: Contemporary Africa and the legacy of late colonialism*. Princeton University Press.
- McGinnis, M. D. (2011). An introduction to IAD and the language of the Ostrom workshop. *Policy Studies Journal*, 39(1), 169–183.
- Mose, L. O., and Ochieng, J. (2021). Devolution and the Implementation of Sustainable Development Goal 2 in Kenya: Addressing the Silo Challenge in County Governments. *Journal of African Development Studies*, 8(2).

- Mwangi, M. N., Bhardwaj, S., and Karanja, S. N. (2021). Institutional challenges to policy implementation: The case of food and nutrition security in Kenya. *Food Policy*, 103, 102040.
- NDMA. (2022). Annual report on drought resilience and response. NDMA.
- Nguyen, P. H., Avula, R., Heidkamp, R. A., Frongillo, E. A., Menon, P., and Kim, S. S. (2017). Maternal and child nutrition in Kenya: A comprehensive review of the literature and recommendations for interventions. *Maternal and Child Nutrition*, 13(3), e12384.
- Ostrom, E. (2005). *Understanding institutional diversity*. Princeton University Press.
- Owuor, J., and Kioli, F. (2017). Devolution and food security in Kenya: A review of literature. *Journal of Food Security*, 5(5), 147–155.
- Pfeffer, J., and Salancik, G. R. (1978). *The external control of organizations: A resource dependence perspective*. Harper and Row.
- Pomeroy-Stevens, A., D'Agostino, A., and Adero, N. (2016). Assessing the enabling environment for nutrition in Kenya: A system-wide analysis. SPRING Project.
- Tegemeo Institute. (2019). Market access and post-harvest losses for Kenyan maize farmers. Tegemeo Working Paper.
- Tegemeo Institute. (2020). The impact of inter-ministerial coordination on agricultural project outcomes in Kenya.
- UNDESA. (2022). *The sustainable development goals report 2022*. United Nations.
- Weber, M. (1922). *Economy and society*. University of California Press.
- World Bank. (2019). *Breaking down silos: A guide to integrated agricultural development in Sub-Saharan Africa*.